

**KWAME NKRUMAH UNIVERSITY OF SCIENCE AND
TECHNOLOGY, KUMASI
COLLEGE OF SCIENCE
DEPARTMENT OF STATISTICS AND ACTUARIAL SCIENCE**



**FACTORS INFLUENCING STUDENT ACADEMIC
PERFORMANCE IN KNUST**

By

GROUP TOKYO

**A THESIS SUBMITTED TO THE DEPARTMENT OF STATISTICS AND
ACTUARIAL SCIENCE, KWAME NKRUMAH UNIVERSITY OF SCIENCE
AND TECHNOLOGY, KUMASI, IN PARTIAL FULFILLMENT OF THE
REQUIREMENT FOR THE COURSE**

OF

**SCIENTIFIC RESEARCH AND COMMUNICATION
(STATISTICS III)**

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Declaration

We affirm that this thesis represents the culmination of our independent and original research efforts, and that no portion of it has been previously submitted for assessment or included in another thesis at this university or any other academic institution.

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Dedication

We humbly dedicate this work to the Almighty God, acknowledging His unwavering strength and boundless mercy that have guided us throughout this journey.

Abstract

In this study, we delved into the complex realm of factors influencing academic performance among students at KNUST. Our exploration encompassed a variety of dimensions, including class size, study hours, socioeconomic background, age, and the relationship between these elements and academic success. Utilizing a blend of quantitative and qualitative methods, we collected and analyzed data from a diverse sample of students. The findings underscored the intricate interplay of socio-economic, psychological, and environmental factors in shaping academic achievements. Notably, the correlation between study hours and academic performance emerged as weakly positive, indicating that dedicating more time to studying could lead to slight improvements in academic outcomes. Our regression analysis highlighted a statistically significant connection between study hours and academic performance, suggesting that increased study time could indeed contribute to better results. However, the model's predictive power remained limited, as reflected by relatively low R-squared values. These findings have implications for educational policies and practices, emphasizing the importance of holistic approaches that consider various factors in fostering student success. As we navigate the complexities of academic pursuits, this study offers insights to guide future endeavors and enhance educational experiences.

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Chapter 1

INTRODUCTION

1.1 Introduction

This Chapter contains the background, purpose, objectives, questions, hypotheses and significance of the study.

1.2 Background of the Study

Academic performance is a crucial aspect of a student's educational journey and is influenced by various factors, including demographic and socioeconomic elements. Demographic factors, such as age and gender, along with socioeconomic status, significantly contribute to a student's academic achievements. Socioeconomic status is determined by an individual's achievements in education, employment, occupation, and income, and it plays a pivotal role in shaping a student's academic performance.

This study focuses on understanding the impact of socioeconomic and demographic factors on the academic performance of students. Socioeconomic status, characterized by family income, class size, age, and study hours, serves as a critical domain in influencing students' educational outcomes. Research has consistently shown that students from higher socioeconomic backgrounds tend to perform better academically compared to students from lower socioeconomic backgrounds.

In recent times, the issue of unpaid school fees has emerged as a significant challenge affecting students' academic performance at Kwame Nkrumah University of Science and Technology (KNUST). Last year, a staggering number of 6000 students faced deferment due to unpaid school fees, and this year, a

majority of students are still struggling to fully pay their fees. Such financial constraints severely impact students' ability to focus on their studies, leading to increased stress and hindered academic performance. Imagine a student being deferred from writing exams simply because they cannot afford to pay their fees. Given the critical nature of the problem, it is imperative to delve deeper into the role of socioeconomic and demographic factors in determining student academic performance at KNUST. By understanding the complexities of these factors, appropriate interventions and support systems can be devised to aid students in overcoming financial challenges and improve their overall educational outcomes. This study aims to investigate the interplay between socioeconomic and demographic factors and their effects on the academic performance of students at KNUST. The findings will contribute valuable insights to the educational community, enabling the implementation of targeted strategies to enhance student success and ensure a conducive learning environment for all. Ultimately, bridging the gap between socioeconomic disparities will pave the way for a more equitable and thriving educational experience for every student at KNUST.

1.3 Problem Statement

Academic performance, which is measured by examination results, is one of the major goals of our school. As an institution aspiring to be a center of excellence, we are deeply committed to quality assurance and maintaining high standards. However, our academic administrators and the quality assurance committee have observed that while some students excel academically, others do not perform as well. This discrepancy in performance raises concern, as it may impact our school's reputation and the confidence in our graduates.

Despite this concern, there is a lack of sufficient research on the specific factors influencing academic performance among our undergraduate students. Therefore, this research project aims to identify and examine the factors affecting academic

performance of undergraduate students at our school, with a specific focus on class size, socio-economic status, and study hours. By investigating these factors, we aim to gain valuable insights that can help us enhance student success and uphold our vision of becoming a center of academic excellence.

1.4 Purpose of the Study

While some undergraduate students at KNUST excel academically, others may face challenges in attaining desired academic standards. The factors contributing to these differences in performance are not yet fully understood. This research seeks to address this gap by exploring the potential influence of class size, socioeconomic status, and study hours on undergraduate students' academic performance at KNUST.

1.5 Objective of the Study

1. To examine the relationship between class size and undergraduate students' academic performance at KNUST.
2. To assess how socioeconomic status influences academic achievement among undergraduate students.
3. To explore the impact of study hours on the academic performance of undergraduate students at KNUST.

1.6 Research Questions

1. Research Question 1: What is the relationship between class size and academic performance among undergraduate students at KNUST?
2. Research Question 2: How does socioeconomic status influence the academic achievement of undergraduate students at KNUST?

3. Research Question 3: What is the impact of study hours on the academic performance of undergraduate students at KNUST?

1.7 Significance of the Study

Understanding the relationship between class size, socioeconomic status, study hours, and academic performance is essential for KNUST's continuous improvement strategies. The findings will offer valuable insights that can aid in tailoring teaching methodologies, support services, and institutional policies to better meet the needs of undergraduate students. Moreover, the study's results will contribute to the existing body of knowledge on factors influencing academic performance in higher education institutions.

1.8 Scope and the Delimitation of the Study

The study will specifically examine the relationships between class size, socioeconomic status, and study hours with academic achievement among undergraduate students.

The target population for this study comprises undergraduate students currently enrolled at KNUST. The research will aim to include a diverse representation of students from various faculties, departments, and academic backgrounds within the university.

A representative sample of undergraduate students will be selected using stratified random sampling to ensure a fair representation from different colleges and departments at KNUST. The final sample size will be determined based on feasibility and statistical considerations.

While this study seeks to provide valuable insights into the factors influencing academic performance among undergraduate students at KNUST, it is important to acknowledge certain limitations that may impact the generalization of the findings. The delimitation include:

1. **Sample Size:** The study's findings will be based on a selected sample of undergraduate students at KNUST, and the results may not represent the entire student population.
2. **Time Constraints:** The study is conducted within a specific time frame, which may limit the depth of analysis or exploration of long-term trends.
3. **Contextual Factors:** The research focuses on undergraduate students at KNUST and may not directly apply to students in other academic institutions or different cultural and regional contexts.
4. **Data Collection Method:** The study will utilize quantitative data collected through surveys and questionnaires, which may limit the depth of understanding that qualitative approaches can provide.
5. **External Variables:** While the study aims to examine the impact of class size, socioeconomic status, and age on academic performance, other external factors not included in the research may also influence student achievement.

By acknowledging these delimitation, the study maintains a clear focus on its objectives and provides valuable insights into the factors affecting academic performance among undergraduate students at KNUST. The results can inform institutional strategies and contribute to the broader body of knowledge on student success in higher education.

1.9 Operational Definition of Terms

Socio-Economic Background: Socio-economic background refers to the social and economic status of a student's family or household, including factors such as income, occupation, education level, and living conditions.

Class Size: Class size refers to the number of students enrolled in a particular course or educational program. It is a measure of the size of the student cohort in a given academic setting.

Study Hours: Study hours represent the amount of time a student dedicates to studying academic materials outside of formal classroom hours. It includes activities such as reading, research, and preparation for assessments.

Continuous Weighted Average (CWA): CWA is a metric used to quantify a student's overall academic performance by considering the grades earned across multiple courses. It reflects the average of the weighted grades based on credit hours or unit values.

Regression Analysis: Regression analysis is a statistical technique used to examine the relationship between one or more independent variables (such as study hours) and a dependent variable (such as academic performance) to understand the extent to which the independent variables predict changes in the dependent variable.

Correlation: Correlation refers to the statistical measure of the strength and direction of a linear relationship between two quantitative variables, such as age and academic performance or study hours and academic performance.

Chapter 2

LITERATURE REVIEW

2.1 Introduction

The literature review pertaining to this chapter is related to factors which influence academic performance among undergraduate student in KNUST. It outlines the literature relation to the relationship between class size and academic performance among undergraduate in KNUST, how socioeconomic status of guardians influence the academic achievement of the undergraduate in KNUST, the impact of study hours on the academic performance affecting undergraduate student in KNUST, summary of the reviewed literature, theoretical and conceptual framework.

2.2 Theoretical Literature Review

Academic performance is a complex outcome influenced by a range of factors encompassing both internal and external dimensions. This section presents a comprehensive review of theoretical perspectives that highlight the various factors that contribute to student academic performance within the context of Kwame Nkrumah University of Science and Technology (KNUST).

2.2.1 Socio-Economic Factors

Socio-economic status (SES) has been recognized as a key determinant of academic performance (Aryee, 2015). Research suggests that students from different socio-economic backgrounds may experience varying levels of access to educational resources and support systems, ultimately impacting their academic outcomes (Aryee and Asante, 2016). SES can influence factors such as access to

learning materials, quality of housing, and access to technology, all of which can influence a student's ability to perform well academically (Chowa et al., 2014).

2.2.2 Study Habits and Time Management

Effective study habits and time management practices are vital for academic success. Research has shown that students who allocate adequate time for studying, employ active learning strategies, and maintain consistent study routines tend to perform better academically (Kumar, 2015). Time management skills are particularly relevant in higher education settings where students are required to balance academic responsibilities with other commitments (Britton and Tesser, 1991).

2.2.3 Environmental Factors

The learning environment, including class size, has been a subject of scholarly attention due to its potential impact on student academic performance. Research has indicated that smaller class sizes can lead to improved teacher-student interactions and greater individual attention (Blatchford et al., 2011). Additionally, the availability of supportive learning environments, such as well-equipped libraries and study spaces, can positively influence student learning experiences (Ko, 2018).

2.3 Empirical Review

Several studies have investigated the factors that influence the academic performance of students in various educational settings. This section presents an empirical review of research specifically related to academic performance factors among students at the Kwame Nkrumah University of Science and Technology (KNUST).

2.3.1 Influence of Class Size on Academic Performance

Research has shown a significant relationship between class size and academic performance. Smith and Johnson (2015) conducted a study at KNUST and found that students in smaller classes tend to have higher academic achievement compared to those in larger classes. Similarly, Brown et al. (2018) found that smaller class sizes allow for more individualized attention, leading to better engagement and improved learning outcomes. Research conducted by education psychologist like Glass and Smilk(1979) suggested that smaller class size could lead to improved student engagement, personalized instruction and a good interaction between the lecturer and student ultimately enhancing academic outcomes. This was supported by studies such as Tennessee STAR project (Word et al., 1990), which reported positive effect on smaller class on student achievement. However, other researches like Hanushelc(1997) argued that while smaller class size must be beneficial, the effect size is often smaller than anticipated and can be largely influence by factors like teaching quality. They emphasized that teacher equality and instructional practices play a vital role in increasing the relationship between class size and academic performance. Betls and Shkolnik(1999) pound that smaller class size in colleges settings have a positive impact on students' course grades. The impact of class size may vary across different educational levels and settings, suggesting a nuanced approach is necessary when interpreting and applying the findings of these study.

2.3.2 Influence of Study Hours on Academic Performance

The relationship between study hours and academic performance has been widely explored. A study by Williams and Lee (2017) at KNUST revealed that students who dedicate more time to studying generally achieve higher academic performance. Conversely, Anderson and Davis (2016) suggested that excessive study hours without breaks can lead to burnout and negatively impact academic performance.

2.3.3 Influence of Socio-economic Background on Academic Performance

A combination of factors which influence students' academic performance has been mentioned. On learning resources, it was found that class size has an impact on students' academic performance. It has also established that there is a relationship between socioeconomic status and academic achievement of students' performance. A study conducted by Adu et al. (2019) found that students from higher socio-economic backgrounds tend to have better access to resources such as private tutoring and educational materials, which positively influence their academic performance. On the other hand, students from low-income backgrounds may face challenges related to financial constraints and lack of resources (Johnson and Smith, 2018), potentially affecting their academic achievements.

2.4 Conceptual Framework

The conceptual framework for this study is based on a comprehensive exploration of the various factors that influence academic performance among students at KNUST. This framework is designed to provide a structured understanding of the interplay between different elements and their potential impact on student success.

Independent Variables:

Class Size: The number of students in a class can affect the level of individual attention, interaction with instructors, and collaborative learning opportunities.

Study Hours: The amount of time students dedicate to studying directly relates to their engagement with course material and depth of understanding.

Socioeconomic Status: Family income and socioeconomic background can influence access to resources, learning environment, and academic support.

Dependent Variable:

Academic Performance (CWA): The Continuous Weighted Average (CWA) serves as an indicator of students' overall academic achievement.

Contextual Factors:

University Policies: Institutional policies on class allocation, study spaces, and financial assistance programs can influence the dynamics between independent and dependent variables.

Curriculum: Course structure, teaching methodologies, and assessment strategies can impact students' study habits and performance.

2.5 Summary

This chapter serves as a synthesis of relevant literature, helping readers grasp the complex web of factors that influence academic performance. The insights gained from this literature review will guide your study's focus, research questions, and methodology in investigating the factors affecting students' academic performance at KNUST.

Chapter 3

RESEARCH METHODOLOGY

3.1 Introduction

In this chapter, we present the methodology used to examine the diverse factors influencing undergraduate student performance at KNUST (Kwame Nkrumah University of Science and Technology). The aim is to offer a detailed account of the research design, the target population under study, the sample size and sampling technique employed, data collection procedures, data analysis methods, as well as the ethical considerations taken into account throughout the study. By elucidating these elements, we ensure a robust and systematic approach to our investigation of the factors that impact student academic performance at KNUST.

3.2 Research Design

This study adopts a correlational research design, employing quantitative data collection methods, with the primary objective of investigating the factors that influence undergraduate student performance at KNUST (Kwame Nkrumah University of Science and Technology). The study aims to explore and establish the relationships between several factors and academic performance. These factors include socioeconomic status, age, class size, and study hours. By analyzing the data collected on these variables, the study seeks to identify their respective impacts on student academic performance at KNUST.

3.3 The Target Population

The target population for this study comprises undergraduate students currently enrolled at KNUST (Kwame Nkrumah University of Science and Technology). The research aims to ensure a comprehensive representation of undergraduate students from various colleges, faculties, and departments within the university. The KNUST enrollment report indicates a student population exceeding 100,000, making this vast student body the focus of our investigation. As the study is specifically focused on factors influencing academic performance, the respondents selected for participation were exclusively undergraduate students, as they constitute the primary demographic relevant to the research objectives at KNUST.

3.4 Sample Size and Sampling Technique

The sample for this study consists of 600 undergraduate students, selected from the extensive student population of over 100,000 at KNUST (Kwame Nkrumah University of Science and Technology). These 600 respondents were drawn from diverse academic backgrounds, representing various colleges within the university. Specifically, participants were selected from the College of Science, College of Engineering, College of Art and Built Environment, College of Health Sciences, College of Humanities and Social Sciences, and College of Agriculture and Natural Resources.

During the data collection process, 578 out of the 600 selected students completed and returned the questionnaires. The study employed a stratified sampling technique to ensure a representative sample. This involved dividing the target population into distinct strata based on the colleges and then applying a simple random sampling approach to select participants from each college. By using this sampling technique, the study aimed to gather valuable insights from a diverse group of undergraduate students across different academic disciplines at KNUST.

3.5 Data Collection

Data for this study will be collected through a structured questionnaire designed to explore the different factors influencing students' academic performance. The questionnaire will encompass variables related to socioeconomic status, such as parents' educational background and salary. Additionally, demographic information, including age and gender, will be gathered to provide a comprehensive understanding of the participants' characteristics.

To facilitate efficient data collection, an electronic-based questionnaire will be utilized, prepared using Google Forms. This approach was chosen for its user-friendly nature, enabling easy completion by respondents while saving time and ensuring their focus remains on the subject matter. The questionnaire will serve as the primary method of data collection, allowing for a systematic and standardized approach in gathering information from the participants.

3.6 Data Analysis

To explore the influence of class size and socioeconomic status on academic performance (CWA), the Kruskal-Wallis rank sum test is applied. This non-parametric test is chosen because the CWA data does not exhibit a normal distribution. The Kruskal-Wallis test allows us to determine if there are significant differences in CWA across different levels of class size and socioeconomic status. By using this test, the study aims to identify any potential associations between these variables and academic performance that may not be captured by parametric tests.

Furthermore, the relationships between age and CWA, as well as study hours and CWA, are examined using the Pearson correlation coefficient. This method assesses the strength and direction of linear relationships between two continuous variables. By calculating the Pearson correlation coefficient, the study seeks to determine whether there is a significant correlation between age, study hours,

and academic performance. This analysis provides insights into whether these variables are positively, negatively, or neutrally associated with CWA.

Additionally, the section employs regression analysis to investigate the relationship between study hours and CWA. Linear regression is used to model the potential linear association between these two variables. The regression analysis helps in understanding how changes in study hours may predict variations in academic performance. This method offers insights into the extent to which study hours contribute to changes in CWA and allows for the identification of any statistically significant relationship.

3.7 Ethical Consideration

The management of participants' academic performance lies within the purview of the University. Therefore, prior authorization was obtained from the Vice Chancellor to carry out this research. Stringent measures were undertaken to ensure the confidentiality of the respondents' provided information, given the exclusive academic focus of the study. To uphold this confidentiality, all information used in the research was attributed with anonymous codes, thereby eliminating any possible linkage to individual students.

3.8 Summary

To achieve the research objectives, a quantitative approach will be employed. Data will be collected through surveys and questionnaires distributed to a representative sample of undergraduate students at KNUST. The sample will be selected using stratified random sampling, ensuring a diverse representation of students from various faculties, departments, and backgrounds. Statistical analysis will be conducted to explore the relationships between class size, socioeconomic status, age, and academic performance.

Chapter 4

DATA PRESENTATION AND ANALYSIS

4.1 Introduction

In this chapter we will use both numbers and descriptions to show the interesting things we discovered. We asked lots of students from different colleges of the university to answer questions, and we looked at their answers closely. By doing this, we want to understand the main reasons why some students do better in their studies compared to others.

We'll use graphs and explanations to help you see how different things are connected, like age, family background, and how much time students spend studying are linked to their academic performance.

4.2 Respondents by gender

Table 4.1: Breakdown of Participants by Gender

Gender	Frequency	Percent
Male	276	47.8
Female	302	52.2
Total	578	100

Table 4.1 provides an overview of the participants, totaling 578 respondents. Among them, female students constituted the majority, accounting for approximately 52.2% of the total. On the other hand, male respondents made up a slightly smaller proportion, comprising 47.8%. This distribution can be

attributed to the higher level of cooperation exhibited by the female participants compared to their male counterparts.

4.3 Respondents by Age

Table 4.2: Participants' Age Distribution

Age	18	19	20	21	22	23	24
Frequency	20	46	59	103	72	101	64
Percent	11.07	3.46	7.96	10.21	17.82	12.46	17.47
Age	25	26	27	28	29	30	
Frequency	52	31	21	5	3	1	
Percent	9.0	5.36	3.63	0.87	0.52	0.17	

Table 4.2 reveals that the participants' ages range from 18 to 30 years. Among the age groups, those aged 22 constitute the largest segment, accounting for approximately 17.82% of the total respondents. Additionally, participants aged 21 and 23 also form significant portions of the sample, with percentages of 10.21% and 12.46%, respectively.

As the age increases beyond 23, the percentages gradually decrease. For instance, respondents aged 25 and 26 represent 9.0% and 5.36% of the total, respectively. Moreover, the oldest participants in the study, aged 30, have a lower representation with a percentage of 0.17%.

Overall, the table provides insights into the age distribution of the participants, highlighting the varying proportions of respondents across different age groups. This information is crucial for understanding the demographic composition of the study's sample and its potential implications for the factors affecting academic performance.

4.4 Respondents by Colleges

Table 4.3: Distribution of Respondents by Colleges of Study

Colleges	Frequency	Percent
Agriculture and Natural Resources	108	18.69
Art and Built Environment	111	19.20
Engineering	65	11.25
Health Sciences	80	13.84
Humanities and Social Sciences	78	13.49
Science	136	23.53
Total	578	100

Table 4.3 emphasizes the range of academic interests that participants pursue at KNUST. It is evident that various colleges contribute significantly to the research sample. Notably, the College of Science holds the highest representation among the respondents, reflecting its importance as a hub for scientific and research-oriented disciplines.

Moreover, the table reveals that College of Agriculture and Natural Resources, College of Art and Built Environment, College of Engineering, College of Health Sciences, and College of Humanities and Social Sciences all make substantial contributions to the study. This distribution highlights that students from various academic fields are participating, showing that the research involves different areas of study and students are working together from different subjects.

4.5 Parents' social economic status as perceived by the respondents

Table 4.4: Socioeconomic status

Income Status	Frequency	Percent
Low-income	125	21.63
Middle-income	277	47.92
High-income	149	25.78
Not sure	27	4.67
Total	578	100

Table 4.4 reveals that out of the total 578 respondents, 125 participants (21.63%) fall under the category of low-income, while the majority of participants, 277 (47.92%), belong to the middle-income group. Additionally, 149 participants (25.78%) are categorized as high-income, and a smaller portion of 27 participants (4.67%) were not sure about their income status.

The distribution provides insight into the diverse socioeconomic backgrounds of the participants, reflecting the representation of various income levels within the study sample. This information is valuable for understanding how different socioeconomic statuses might relate to other variables and factors being investigated in the study.

4.6 Respondents Class sizes

Table 4.5: Distribution of Respondents by Class Size

Class Size	Frequency	Percent
Less than 100	41	7.10
101 - 200	210	36.33
201 - 300	273	47.23
More than 300	54	9.34
Total	578	100

From the table 4.5, it can be observed that out of the total 578 respondents, 41 participants (7.10%) are enrolled in classes with less than 100 students. The largest portion of participants, 210 (36.33%), are part of classes with 101 to 200 students. Moreover, 273 participants (47.23%) belong to classes with 201 to 300 students, and a smaller group of 54 participants (9.34%) are enrolled in classes with more than 300 students.

This distribution provides valuable insights into the diversity of class sizes experienced by the participants. It highlights the representation of different class size categories within the study sample and offers context for further analysis of the relationship between class size and academic performance.

4.7 Description of the Dependent Variable (Continuous Weighted Average - CWA)

The dependent variable in this study is Continuous Weighted Average (CWA), which serves as a measure of academic performance. CWA is a numerical representation of a student's overall performance in their courses, calculated by assigning weights to individual course grades based on the credit hours of each

course. The CWA values range from a minimum of 0 to a maximum of 100.

CWA is used as a quantitative indicator to assess the academic achievements of students. Higher CWA scores indicate better academic performance, while lower scores suggest relatively lower performance. CWA takes into account the grades obtained in all the courses a student has taken, providing a comprehensive evaluation of their overall academic success.

In this study, CWA is the focus of investigation to understand how it is influenced by various factors, such as class size, demographic characteristics, and other potential variables. Analyzing the relationship between CWA and these factors can offer insights into the determinants of students' academic achievements and provide valuable information for educational institutions to enhance student performance and learning outcomes.

4.8 Summary of the relationship between Class size and CWA

Table 4.6: Kruskal-Wallis rank sum test on Class size and CWA

Kruskal-Wallis chi-squared	Degrees of Freedom	P-value
4.1008	3	0.2508

The Kruskal-Wallis rank sum test was conducted to examine the potential relationship between Continuous Weighted Average (CWA) and Class Size.

Table 4.6 emphasis that Kruskal-Wallis chi-squared statistic is calculated as 4.1008, and it is associated with degrees of freedom (df) equal to 3. The p-value obtained from the test is 0.2508.

The p-value of 0.2508 is above the conventional significance level (such as 0.05). This suggests that there is insufficient evidence to reject the null hypothesis. In other words, we do not have enough evidence to conclude that there is a

statistically significant difference in CWA across different class sizes.

In summary, based on the Kruskal-Wallis rank sum test, we do not find significant evidence to suggest that class size has a significant impact on Continuous Weighted Average (CWA) in the given data set. This means that the differences in CWA across different class sizes could be due to random variation rather than a meaningful relationship.

4.9 Summary of the association between Socioeconomic status and CWA

Table 4.7: Kruskal-Wallis rank sum test on Family income and CWA

Kruskal-Wallis chi-squared	Degrees of Freedom	P-value
16.826	4	0.002089

The Kruskal-Wallis rank sum test was conducted to assess the potential relationship between the Continuous Weighted Average and Family Income among the study participants. The test evaluates whether there are significant differences in CWA scores across different levels of family income.

Table 4.7 emphasis that Kruskal-Wallis chi-squared statistic obtained was 16.826, with degrees of freedom equal to 4. The associated p-value was calculated to be 0.002089.

The small p-value (0.002089) suggests that there is evidence to reject the null hypothesis, indicating that there are statistically significant differences in CWA scores among different levels of family income. This means that the family income of the participants appears to have an impact on their academic performance, as measured by CWA.

This result highlights the importance of family income as a potential factor influencing students' academic achievements. It suggests that students from

different income backgrounds may have varying CWA scores, implying that economic factors could play a role in shaping academic performance.

4.10 Summary of Impact of Age on CWA

Table 4.8: Correlation between Age and CWA

Pearson correlation coefficient
0.06469826

The correlation coefficient of approximately 0.0647 suggests that there is a very weak positive linear relationship between Age and CWA (Continuous Weighted Average). This implies that as the Age of students increases, there is a minimal tendency for their CWA to also increase. However, the correlation is quite low, indicating that Age alone is not a strong predictor of changes in CWA.

4.11 Summary of Impact of Study hours on CWA

Table 4.9: Correlation between Study hours and CWA

Pearson correlation coefficient
0.1318817

From table 4.9, the correlation coefficient between "Study hours" and "CWA" is approximately 0.132. This value is positive, indicating a positive linear correlation between the two variables. However, the magnitude of the correlation coefficient is relatively low.

This suggests that there is a weak positive linear relationship between the amount of time spent studying (study hours) and academic performance (CWA). In other

words, as study hours increase, there tends to be a slight increase in academic performance, but the relationship is not very strong.

Table 4.10: Regression Results: Impact of Study Hours on CWA

Variable	Coefficient	Standard Error	p-value
Intercept	62.2492	0.7506	$< 2e - 16$
Study Hours	0.1862	0.0583	0.00148

Table 4.11: Model Evaluation Metrics

Metric	Value
Residual Standard Error	8.684
Multiple R-squared	0.01739
Adjusted R-squared	0.01569
F-statistic	10.2
Degrees of Freedom (Model)	1
Degrees of Freedom (Residual)	576
p-value	0.001485

The linear regression model you ran is:

$$CWA = 62.2492 + 0.1862 \times \text{StudyHrs}$$

Interpretation of the coefficients:

Intercept (62.2492):

This is the predicted value of CWA when Study Hours is 0.

Study hours (0.1862):

This coefficient represents the change in the predicted CWA for a one-unit increase in Study Hours.

The p-value associated with this coefficient is 0.00148, which is less than the typical significance level of 0.05. This suggests that there is evidence to conclude that the relationship between Study hours and CWA is statistically significant. In other words, as students increase their study hours, their academic performance

(CWA) tends to improve.

Model goodness of fit:

Residual standard error (8.684):

This is the estimate of the standard deviation of the residuals, which are the differences between the observed and predicted values of CWA. It gives you an idea of how well the model fits the data.

Multiple R-squared (0.01739):

This is the proportion of the variance in the CWA variable that is explained by the Study hours variable. The R-squared value is quite low, indicating that the Study hours variable explains only a small portion of the variation in CWA.

Adjusted R-squared (0.01569):

This is a modified version of R-squared that adjusts for the number of predictors in the model. It's also relatively low in this case. F-statistic (10.2) and p-value (0.001485):

The F-statistic tests the overall significance of the regression model. The p-value indicates whether the model as a whole is statistically significant. In this case, the p-value is less than 0.05, suggesting that the model provides a significant improvement over a constant-only model. In summary, the linear regression analysis suggests that there is a statistically significant relationship between Study hours and academic performance (CWA). As students increase their study hours, their academic performance tends to improve. However, keep in mind that the model's overall ability to predict academic performance based on study hours is limited, as indicated by the relatively low R-squared values.

4.12 Summary

This chapter brings together all the important findings of the study, giving us a clear understanding of how different things affect students' success at KNUST. By carefully looking at the data and explaining what it means, this chapter helps

us understand how various factors work together to impact how well students do in their studies. This understanding can help us make important decisions for the future of education at KNUST.

Chapter 5

DISCUSSION OF THE FINDINGS

5.1 Introduction

In this chapter, we delve into the results outlined in Chapter Four and analyze what they mean. Additionally, we draw conclusions based on these findings and put forward recommendations that have the potential to enhance academic performance at KNUST.

5.2 Discussion of findings

5.2.1 Objective one

Upon analyzing the findings of the first objective, which focused on the relationship between class size and Continuous Weighted Average (CWA), it is evident that the Kruskal-Wallis rank sum test did not yield significant evidence to support the notion that class size has a substantial impact on students' CWA within the provided dataset. This implies that the variations in CWA observed across different class sizes could be attributed to chance fluctuations rather than a meaningful correlation.

5.2.2 Objective two

The findings from the second objective of the study indicate a significant relationship between students' socioeconomic backgrounds and their Continuous Weighted Average (CWA) scores. This implies that students coming from different income backgrounds exhibit varying levels of academic performance. The influence of economic factors on academic success is evident, and it aligns

with existing research that underscores the impact of socio-economic status on educational outcomes.

5.2.3 Objective three

The findings from the analysis of the third objective indicate a weak positive linear relationship between study hours and academic performance, as measured by the Continuous Weighted Average (CWA). This implies that, on average, students who spend more time studying tend to achieve slightly higher academic performance. However, it's important to note that this relationship is not very strong, suggesting that other factors beyond study hours are also at play in determining academic success.

The linear regression analysis conducted further reinforces this notion. The statistically significant relationship between study hours and academic performance highlights the potential impact of dedicating more time to studying. As students increase their study hours, there is a tendency for their academic performance to improve.

5.3 Conclusion

The following conclusions drawn as a result of the research work carried out in the area of academic performance of undergraduate students of KNUST reflect both the theoretical and practical lessons, which can be drawn from the study.

5.3.1 Objective one

The statistical analysis suggests that class size might not be a decisive factor influencing students' academic performance in this context. However, it's important to consider that other unaccounted variables could still play a role in shaping academic outcomes.

5.3.2 Objective two

In conclusion, the study underscores the significance of socioeconomic status in shaping students' academic performance. The results suggest that students from disadvantaged economic backgrounds might face additional challenges that hinder their academic progress. Conversely, those from higher-income backgrounds might have more resources and support that contribute to better academic outcomes.

5.3.3 Objective three

In conclusion, the findings support the idea that study hours can have a modest influence on academic performance. While the relationship is statistically significant, it's important to recognize that study hours alone might not be the sole determinant of strong academic outcomes. Other factors such as learning techniques, subject comprehension, and personal motivation also contribute to academic success.

5.4 Recommendations

5.4.1 Objective one

We recommended that further investigations delve into potential hidden factors that might contribute to variations in academic performance. Additionally, educational institutions could focus on optimizing teaching methodologies and student support systems to ensure consistent academic achievements across varying class sizes. This could involve strategies such as personalized attention, interactive learning environments, and effective utilization of resources to enhance the overall learning experience regardless of class size.

5.4.2 Objective two

Based on these findings, it is recommended that educational institutions, including KNUST, consider implementing targeted support mechanisms for students from lower-income backgrounds. This could involve providing financial aid, mentorship programs, or additional academic resources to ensure that economic disparities do not hinder academic achievement. Furthermore, policymakers should focus on creating an inclusive educational environment that caters to the diverse economic backgrounds of students, thus fostering equal opportunities for academic success.

5.4.3 Objective three

Based on these findings, several recommendations can be made to enhance academic performance:

Balanced Approach to Study: Students should aim for a balanced study routine that combines effective study techniques with appropriate breaks. Simply increasing study hours may not yield significant improvements if the quality of study time is compromised.

Time Management: Encouraging students to manage their time effectively can help them allocate sufficient time for both studying and other activities. This can prevent burnout and improve overall academic performance.

Learning Strategies: Educators should provide guidance on effective learning strategies that can maximize the impact of study hours. Techniques such as active learning, practice tests, and concept mapping can enhance comprehension and retention.

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APPENDIX A

Table 5.1: Questionnaire

FACTORS INFLUENCING STUDENTS' PERFORMANCE IN KNUST

"Factors Influencing Students' Performance in KNUST" is a research topic that focuses on identifying and understanding the various elements that have an impact on students' academic achievements and overall performance. This research aims to explore the different factors that influence students' success in educational settings, covering both internal and external aspects.

* Indicates required question

1. What is your gender? *

Mark only one oval.

- ☐ Male
☐ Female

2. What is your age? *

Example: 22

3. What is the highest level of education completed by your parents or guardians? *

Mark only one oval.

- ☐ No formal education
☐ Primary school
☐ Secondary school
☐ Diploma or vocational training
☐ Bachelor's degree or higher

4. How would you describe your family's socio-economic status?

Mark only one oval.

- ☐ Low-income
- ☐ Middle-income
- ☐ High-income
- ☐ Not sure

FACTORS INFLUENCING STUDENTS' PERFORMANCE IN KNUST

"Factors Influencing Students' Performance in KNUST" is a research topic that focuses on identifying and understanding the various elements that have an impact on students' academic achievements and overall performance. This research aims to explore the different factors that influence students' success in educational settings, covering both internal and external aspects.

5. Please select your College. *

Mark only one oval.

- ☐ College of Science
- ☐ College of Engineering
- ☐ College of Agriculture & Natural Resources
- ☐ College of Art and Built Environment
- ☐ College of Health Sciences
- ☐ College of Humanities & Social Sciences

6. What your current Continuous Weighted Average (CWA). *

Example: 66.77

7. Please select your class size. *

Mark only one oval.

☐ Less than 100 Students

☐ 101 - 200 Students

☐ 201 - 300 Students

☐ More than 301 Students

8. On average, how many hours do you spend studying per week? *

Example: 10 hours

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